

Call for Expressions of Interest	
<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	Synthetic blades for 3D reconstruction
<i>Company:</i>	University of Nottingham
Contact Details	
<i>Lead Contact:</i>	Dr. Yancong Lin
<i>Lead E-mail:</i>	yancong.lin@nottingham.ac.uk
<i>Additional Contact:</i>	Additional contact name
<i>E-mail:</i>	Additional contact e-mail
<i>Additional Contact:</i>	Additional contact name
<i>E-mail:</i>	Additional contact e-mail
<i>Additional Contact:</i>	Additional contact name
<i>E-mail:</i>	Additional contact e-mail
Additional Info	
<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☐
<i>Custom hardware required:</i>	☒ If Applicable: GPUs
<i>Relevant research areas (if applicable)¹:</i>	Computer Vision, Computer Graphics
Project Background Information	
Recent advances in deep learning have significantly improved the ability to reconstruct high-quality 3D models from collections of images, with methods such as Neural Radiance Fields (NeRF) and 3D Gaussian Splatting achieving impressive results. However, these techniques are typically developed and evaluated under ideal conditions, assuming well-textured objects, consistent illumination, and uncluttered environments. In industrial applications, these assumptions rarely hold. Components such as aircraft engine blades often exhibit texture-less and highly reflective surfaces, repetitive geometric patterns, and challenging lighting conditions, making accurate 3D reconstruction considerably more difficult. This project aims to investigate the limitations of state-of-the-art 3D reconstruction methods in realistic industrial scenarios and explore strategies to improve their robustness through the use of synthetic data. The project will involve creating realistic 3D models of aircraft engine blades in Blender and rendering large-scale synthetic datasets consisting of RGB and depth images captured from diverse viewpoints and object poses. The rendering pipeline will simulate a wide range of real-world imaging conditions by varying lighting, backgrounds, and other environmental factors to produce representative training and evaluation data.	

¹ An overview of different research areas in the school of computer science can be found [here](#).

The generated dataset will be used to train and benchmark modern 3D reconstruction algorithms, including NeRF- and Gaussian Splatting-based approaches. By systematically evaluating reconstruction quality under challenging industrial conditions, the project seeks to identify the key factors limiting current methods and assess how synthetic data generation can improve their ability to accurately reconstruct complex engineering components.

Example inspection video: https://www.youtube.com/watch?v=9b4ER1O_PNA

References

<https://www.blender.org/>

<https://colmap.github.io/>

<https://robust-equivariant-sfm.github.io/>

<https://docs.nerf.studio/> and <https://www.matthewtancik.com/nerf>

<https://github.com/graphdeco-inria/gaussian-splatting>

Requirements

The group must provide:

A Blender model for synthetic data generation

An automated pipeline to synthesize data (e.g., metadata, images, camera poses, lighting, etc.)

Skills

Highly desirable:

Programming (Python or C++)

Desirable:

Blender

Submission / Award information

Date of Submission:

September 2026

Date of Pitch:

October 2026

Notification of award:

October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.